

BEGINNER'S GUIDE: YOUR FIRST AUTONOMOUS AI

Step-by-Step Friendly Introduction to Building Autonomous AI Agents with Sapphire

1. Welcome! What is an Autonomous AI?

Welcome to the world of **Autonomous AI Programming**! If you are new to AI development, don't worry. An **Autonomous AI Agent** is simply a computer program that can observe what is happening on your system, think about what needs to be done using AI, and take action automatically.

Analogy: Chatbot vs Autonomous Agent

- **Chatbot**: Like a helpful friend who answers questions when you text them.
- **Autonomous Agent**: Like a personal assistant who checks your PC, fixes problems, sends you notifications, and works 24/7 in the background without being asked!

2. The 3 Steps of Every Autonomous Bot

Every autonomous bot in Sapphire follows 3 simple steps:

Step 1: Check System (Perception) — The bot reads system RAM, disk, or files using `os.system_info()`.

Step 2: Ask AI (Intelligence) — The bot sends telemetry to `ai.prompt(...)` for decision making.

Step 3: Perform Action (Execution) — The bot posts desktop alerts with `os.notify()` or writes files.

3. Writing Your Very First 5-Line AI Agent

Here is the simplest complete Autonomous AI Agent written in Sapphire:

```
// 5-Line Autonomous AI Agent in Sapphire
let info = os.system_info();
let question = "System CPU is at {info.cpu_usage_percent}%. Is this good?";
let ai_opinion = ai.prompt(question);

os.notify("Sapphire Beginner Bot", ai_opinion);
print("■ Agent Output: {ai_opinion}");
```

4. Step-by-Step Code Walkthrough

Code Line	What It Does (Beginner Friendly)
<code>let info = os.system_info();</code>	Gets live CPU and RAM usage from your computer.
<code>let question = "...";</code>	Creates a prompt sentence with your CPU percentage.
<code>let ai_opinion = ai.prompt(question);</code>	Passes the question to Sapphire's native AI engine.
<code>os.notify("...");</code>	Displays a pop-up toast notification on your Windows desktop!
<code>print("...");</code>	Prints the AI answer to your screen.

5. How to Run Your Agent

You can run your Sapphire AI agent in 2 easy ways:

Option A (Interactive Tutor): Double-click `sapphire_tutor.bat` or `sapphire_tutor.exe` and choose option 4 to generate sample code.

Option B (Command Line): Open terminal and type `python sapphire_lang/src/cli.py run my_first_sapphire_bot.sp`.

■ Congratulations!

You have taken your first step into Autonomous AI programming with Sapphire! Try customizing the prompt or adding ``scheduler.interval`` to run your bot automatically every minute.